

# FINAL RESULTS ON FUTURE TEST SET

| Model                     | PR-AUC | ROC-AUC | F2    | MCC   |
|---------------------------|--------|---------|-------|-------|
| Recency-Weighted LightGBM | 0.606  | 0.928   | 0.558 | 0.582 |
| LightGBM Bagging Ensemble | 0.604  | 0.930   | 0.569 | 0.581 |
| MLP (DL Benchmark)        | 0.584  | 0.781   | 0.620 | 0.623 |
| TabNet (DL Benchmark)     | 0.561  | 0.822   | 0.654 | 0.610 |

Conclusion: Recency-Weighted LightGBM - highest PR-AUC (0.606) | Bagging Ensemble - best stability (ROC-AUC 0.930, F2 0.569) | Deep Learning models underperformed